

Increased risk of recurrence and disease-specific death following delayed postoperative radiation for Merkel cell carcinoma



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Background: Merkel cell carcinoma (MCC) is often treated with surgery and postoperative radiation therapy (PORT). The optimal time to initiate PORT (Time-to-PORT [ttPORT]) is unknown.

Purpose: We assessed if delays in ttPORT were associated with inferior outcomes.

Methods: Competing risk regression was used to evaluate associations between ttPORT and locoregional recurrence (LRR) for patients with stage I/II MCC in a prospective registry and adjust for covariates. Distant metastasis and death were competing risks.

Results: The cohort included 124 patients with median ttPORT of 41 days (range: 8-125 days). Median follow-up was 55 months. 17 (14%) patients experienced a LRR, 14 (82%) of which arose outside the radiation field. LRR at 5 years was increased for ttPORT >8 weeks vs ≤ 8 weeks, 28.0% vs 9.2%, $P = .006$. There was an increase in the cumulative incidence of MCC-specific death with increasing ttPORT (HR = 1.14 per 1-week increase, $P = .016$).

Limitations: The relatively low number of LRRs limited the extent of our multivariable analyses.

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Conclusions: Delay of PORT was associated with increased LRR, usually beyond the radiation field. This is consistent with the tendency of MCC to spread quickly via lymphatics. Initiation of PORT within 8 weeks was associated with improved locoregional control and MCC-specific survival. (J Am Acad Dermatol 2024;90:261-8.)

Key words: adjuvant; carcinoma; dermatologic surgical procedures; Merkel Cell; patient outcome assessment; radiotherapy; sentinel lymph node biopsy; skin neoplasms; wound healing.

INTRODUCTION

Merkel cell carcinoma (MCC) is a rare and aggressive neuroendocrine skin cancer with an incidence of approximately 3000 cases a year in the United States, and is rapidly increasing.¹⁻³ At diagnosis, approximately two-thirds of patients with MCC present with localized disease without evidence of regional lymph node (LN) involvement or metastatic disease.^{4,5} Following definitive surgery and pathological nodal evaluation by sentinel LN biopsy, postoperative radiation therapy (PORT) is often recommended to optimize locoregional control. The benefits of PORT on locoregional control for MCC are well documented in the literature, showing an absolute risk reduction in these events by approximately 20%-30%.⁶⁻¹¹

However, a key variable that has not yet been well characterized in MCC is the optimal time interval between surgery and the initiation of PORT (Time-to-PORT [ttPORT]). The importance of timely initiation of PORT has been documented in other malignancies, such as head and neck (H&N) squamous cell carcinoma where delay in the initiation of PORT beyond 6 weeks is associated with worse locoregional control.¹²⁻¹⁶ To address this gap in knowledge, we utilized our prospective MCC registry to evaluate the effect of ttPORT on MCC locoregional recurrence (LRR) rates and other related outcomes.

MATERIALS AND METHODS

MCC Analysis cohort

We retrospectively identified a cohort of 124 patients with localized stage I/II MCC (AJCC eighth ed.)⁴ from our longitudinal prospective registry of pathologically confirmed MCC ($n = 1374$) treated between October 2004 and September 2018. A diagram of the inclusion criteria and exclusion criteria is presented in Supplementary Fig 1, available via

CAPSULE SUMMARY

- For localized Merkel cell carcinoma, the optimal time to initiate postoperative radiation therapy following surgery is unknown.
- Delay of postoperative radiation therapy >8 weeks is associated with increased locoregional recurrence. Most locoregional recurrences occurred outside the radiation field, suggesting that timely initiation of postoperative radiation therapy may impede spread beyond the surgical bed.

Mendeley at <https://doi.org/10.17632/9k9gfx483c.1>. Patients were treated across multiple institutions and categorized as institution 1 versus outside institutions.

Radiation therapy

PORT was delivered using electrons with bolus or photons to the entire surgical bed plus a 3-5 cm margin as permitted by the anatomical location of the primary site. Inclusion of the in-transit lymphatics and/or the elective nodal basin in the PORT fields was at the provider's discretion after an informed discussion based on the location of the primary (H&N vs other) and the successful completion pathologic staging.

Endpoints

The primary endpoint was LRR. Additional endpoints included distant metastases (DM), MCC-specific death (MCCSD), and overall mortality. LRR included local, in-transit, and regional recurrences. Negative margins were defined as no tumor on the ink. *Local failure* was defined as a recurrence arising within 2.0 cm of the primary excision scar; *in-transit failure* as a recurrence arising > 2 cm from the primary scar but prior to the draining regional LN basin; and *regional failure* as a recurrence within the draining LN bed. *Distant* recurrences were defined as occurring beyond the draining LN basin. Only the first recurrence event (LRR or distant recurrences) for each patient was included in the analysis. *In-field vs out-of-field* PORT recurrences were defined as occurring within or outside of the 50% prescription isodose line, respectively.

Surveillance

Standard-of-care patient surveillance was performed as previously reported.¹⁷

Abbreviations used:

DM:	distant metastases
LN:	lymph node
LRR:	locoregional recurrence
MCC:	Merkel cell carcinoma
MCCSD:	MCC-specific death
PORT:	postoperative radiation therapy
ttPORT:	Time-to-PORT

Statistical analyses

The time interval between date of definitive surgery and initiation of PORT was characterized as time to PORT (ttPORT). For all endpoints, the date of definitive surgery was considered time point zero and the date of initiation of PORT was treated as the entry time into the risk set. Fisher exact test and the Wilcoxon rank-sum test were used to compare baseline clinical and treatment characteristics between ttPORT groups. Spearman's rank correlation was used to assess trends between ttPORT and surgical complications. Cumulative incidence curves were used to estimate the probability of each endpoint over time. LRR, DM, and death were treated as competing risks in the analysis of recurrence endpoints. Other cause death was treated as a competing risk in the analysis of MCCSD while overall mortality was analyzed without competing risks. Fine & Gray competing risk regression models were used to evaluate associations of ttPORT and other baseline variables with LRR and the other endpoints with competing risks. Cox regression models were used to evaluate associations with overall mortality.

The number of LRRs limited the extent of our multivariable analyses. The primary model evaluating the association of ttPORT with LRR was adjusted by one baseline characteristic at a time to assess whether there could be any confounding factors. A *P*-value of <0.05 was considered to be statistically significant. The statistical analysis was performed using R version 3.6 (R Foundation for Statistical Computing, Vienna, Austria). All analyses were performed in accordance with Helsinki principals and approved by our institutional review board (IRB #6585).

RESULTS

Patient and tumor characteristics

A total of 124 patients met inclusion criteria for analysis. Patient, tumor, and treatment characteristics are summarized in Table I. The majority of patients within our cohort were male (58.9%) and the median age at diagnosis was 68 years old (range: 40-87). Most patients had stage I disease (69.4%; *n* = 86), all were node negative (83% pN0, 17% cN0), and 13%

had positive margins following definitive surgery (*n* = 16). Non-H&N primary sites had more pathologically confirmed nodal evaluations than H&N primary sites (94%, *n* = 75/79 vs 64%, *n* = 29/45). No patients received concurrent and/or adjuvant immunotherapy.

The analysis cohort was dichotomized by short ttPORT (≤ 8 weeks; *n* = 93) vs long ttPORT (> 8 weeks; *n* = 31) for comparison of baseline demographics and treatment characteristics (Table I). The choice of an 8 week threshold was based on the analysis of locoregional failures on a week-to-week basis that first resulted in a significant difference at 5 years (Fig 1). There were no statistical differences between the short ttPORT (≤ 8 weeks) vs long ttPORT (> 8 weeks) in terms of sex, age at the time of diagnosis, disease stage, lymphovascular invasion, immunosuppression status, primary tumor size, T-stage, positive margin status following definitive surgery, or pathologic vs clinical staging. The median follow-up time was 55 months (range: 3-156 months) for the entire cohort, with a shorter median follow-up noted in the long (> 8 week) ttPORT cohort (42 vs 59 months, *P* = .002).

PORT Treatment characteristics

PORT treatment characteristics are further summarized in Table I. The median total dose and fractionation of PORT in the entire cohort was 50 Gy (range: 42-67 Gy) in 25 fractions (range: 20-37 fractions). The long ttPORT group (> 8 weeks) was irradiated to a higher median dose of 54 Gy in 27 fractions compared to 50 Gy in 25 fractions in the short ttPORT group (*P* = .004). A greater proportion of patients within the long ttPORT cohort received elective PORT to the draining LN basin compared to the short ttPORT cohort (45.2% vs 19.4% *P* = .008). There were no significant differences in the total duration of PORT (*P* = .29) due to treatment interruptions that potentially reduce the efficacy of PORT, or by the institution at which the PORT was administered, Institution 1 vs outside institution (*P* = .094).

The majority of patients received PORT to the primary site alone (*n* = 92; 74.2%) and 25.8% of patients received PORT to the primary sites and elective draining LN basin (*n* = 32). Patients with a H&N primary site more often received comprehensive nodal irradiation compared to non-H&N sites (48.9% vs 12.7%, *P* < .001). Furthermore, the primary site (H&N vs non-H&N, OR: 4.4, *P* = .002) and the clinical vs pathologic node-negative status (OR: 5.6, *P* = .002) were each independently associated with PORT being given to the draining LN basin (Supplementary Table I, available via Mendeley at <https://doi.org/10.17632/9k9gfx483c.1>).

Table I. Patient, tumor and treatment characteristics dichotomized by Time-to-PORT

Variable	N = 124*	ttPORT ≤8 wk (N = 93)	ttPORT >8 wk (N = 31)	P
Sex				>.99
Male	73 (58.9)	55 (59.1)	18 (58.1)	
Female	51 (41.1)	38 (40.9)	13 (41.9)	
Age at diagnosis, y	68 (40-87)	68 (40-87)	69 (45-86)	.59
Stage, AJCC 8th edition				.21
p-I	73 (58.9)	54 (58.1)	19 (61.3)	
c-I	13 (10.5)	8 (8.6)	5 (16.1)	
p-II	30 (24.2)	26 (28.0)	4 (12.9)	
c-II	8 (6.5)	5 (5.4)	3 (9.7)	
Stage, AJCC 8th edition				.37
p/c-I	86 (69.4)	62 (66.7)	24 (77.4)	
p/c-II	38 (30.6)	31 (33.3)	7 (22.6)	
T-Stage				>.99
T1/2	116 (93.5)	87 (93.5)	29 (93.5)	
T3/4	8 (6.5)	6 (6.5)	2 (6.5)	
Head & neck primary	45 (36.3)	29 (31.2)	16 (51.6)	.053
Immunosuppressed	14 (11.3)	9 (9.7)	5 (16.1)	.34
Path margins				.43
Wide (≥ 1 cm)	74 (59.7)	54 (58.1)	20 (64.5)	
Narrow (0.1-0.99 cm)	34 (27.4)	28 (30.1)	6 (19.4)	
Positive	16 (12.9)	11 (11.8)	5 (16.1)	
Total RT dose, Gy	50 (42-67)	50 (42-67)	54 (50-66)	.004
<50 Gy	2 (1.6)	2 (2.2)	0 (0.0)	
50-60 Gy	101 (81.5)	78 (83.9)	23 (74.2)	
≥ 60 Gy	21 (16.9)	13 (14.0)	8 (25.8)	
No. of RT fractions	25 (20-37)	25 (20-37)	27 (24-33)	.027
PORT dose per fraction, Gy	2.0 (1.5-2.5)	2.0 (1.5-2.5)	2.0 (1.8-2.4)	.50
PORT given to the elective draining LN basin	32 (25.8)	18 (19.4)	14 (45.2)	.008
PORT treating Institution				.094
Institution 1	31 (25.0)	27 (29.0)	4 (12.9)	
Outside Institutions	93 (75.0)	66 (71.0)	27 (87.1)	
Time from diagnosis to definitive surgery, days	27 (0-62)	28 (0-62)	23 (0-51)	.033
Time from definitive surgery to PORT (ttPORT), days	41 (8-125)	36 (8-56)	70 (57-125)	<.001
Total duration of PORT, days	36 (22-71)	36 (24-71)	37 (22-49)	.29
Total follow up from definitive surgery, months	55 (3-156)	59 (3-156)	42 (7-118)	.002

c, Clinical; Gy, gray; LN, lymph node; p, pathologic; PORT, post-operative radiation therapy; RT, radiation therapy; ttPORT, time to PORT.

*Values are No. (%) or median (range).

Patterns of failure

Overall, seventeen (14%) of the 124 patients experienced a LRR as their first event (Table II). The time to LRR ranged from 1.8 to 71 months (median: 10.3 months). Of these 17 events, 14 (82%) occurred outside the radiation field. There was 1 in-field recurrence. There were 5 in-transit recurrences, and all 5 were outside the radiation field. There were 11 events in the regional LNs and 9 of these were out-of-field recurrences. The detailed clinical and pathological characteristics of patients with LRR are provided in Supplementary Table II, available via Mendeley at <https://doi.org/10.17632/9k9gfx483c.1>.

Twelve patients (9.7%) experienced distant metastases. Twenty-one patients died, with 11 deaths attributable to MCC within this cohort.

Time to PORT and impact on risk of recurrence

The median time from diagnosis to definitive surgery was 27 days (range: 0-62 days) and the median time from definitive surgery to ttPORT was 41 days (range: 8-125). There was a significant increasing trend of higher risk of LRR with longer ttPORT, with a hazard ratio of 1.20 per 1-week increase in ttPORT (95% CI: 1.07-1.36, $P = .002$). This trend persisted when clinically staged patients

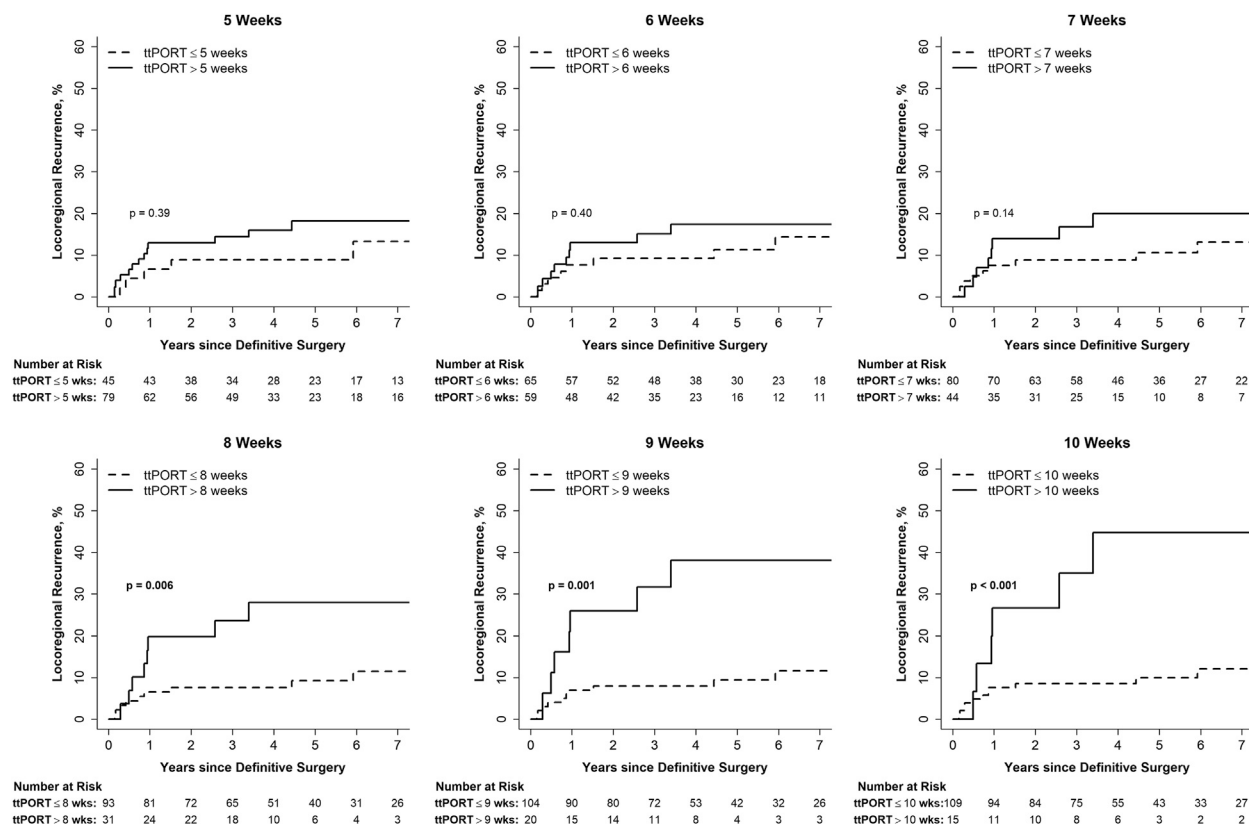


Fig 1. Loco-regional recurrence (LRR) rates, as dichotomized by six increasing time-to-PORT (ttPORT) intervals. LRR rates observed in years from the start of PORT, as the same analysis cohort is dichotomized between ≤ 5 and > 5 (top left), 6 (top middle), 7 (top right), 8 (bottom left), 9 (bottom middle), and 10 (bottom right) weeks of ttPORT. Overall, there was a significant increasing trend toward higher risk of LRR with longer ttPORT, with a hazard ratio of 1.20 per 1-week increase in ttPORT (95% CI: 1.07-1.36, $P = .002$).

Table II. Patterns of failure after PORT among 124 locally staged MCC patients

First event	All (N = 124)	Number of recurrences				
		ttPORT ≤8 wk (N = 93)	ttPORT >8 wk (N = 31)			
Local recurrence	1	0	1			
In-transit recurrence	5	4	1			
Regional recurrence	11	5	6			
Distant metastasis	12	11	1			
Locoregional recurrences in relation to the radiation field	Number in-field	Number out-of-field	Number in-field	Number out-of-field	Number in-field	Number out-of-field
Local	1	0	0	0	1	0
In Transit	0	5	0	4	0	1
Regional	2	9	1	4	1	5
Survival events						
Death from any-cause	21		13		8	
Merkel cell carcinoma-specific death	11		7		4	

ttPORT, Time to postoperative radiation therapy.

Table III. Factors influencing Time-to-PORT

Factor	Total No. (%) (N = 124)	Time-to-PORT		Trend with delay* P
		≤8 wk (N = 93)	>8 wk (N = 31)	
Definitive Surgery, with complication	4 (3.2)	2 (2.2)	2 (6.5)	.088
Skin graft performed, without complication	13 (10.5)	10 (10.8)	3 (9.7)	.84
Skin graft performed, with complication	10 (8.1)	2 (2.2)	8 (25.8)	<.001
Staged surgical procedures [†]	6 (4.8)	2 (2.2)	4 (12.9)	.090

Bold-italic values signify statistical significance.

*Wilcoxon rank-sum test comparing time-to-PORT between those with and without a healing issue or Spearman's rank correlation testing the association between number of healing issues and time-to-PORT.

[†]Definitive surgery and sentinel lymph node biopsy were not performed on the same date.

(did not undergo pathologic nodal evaluation) were removed from the analysis (HR: 1.22 per 1-week increase, 95% CI: 1.07-1.39, $P = .003$) as well as when adjusting for T-stage (HR: 1.20 per 1-week increase, 95% CI: 1.07-1.35, $P = .002$). To further describe this observed relationship, the analysis cohort was sequentially dichotomized by increasing weekly ttPORT thresholds, from 5 to 10 weeks (Fig 1). We observed that LRR rates at 5 years significantly differed starting at ttPORT > 8 weeks as compared to ≤8 weeks (28.0% vs 9.2%, $P = .006$). The ttPORT to LRR relationship continued to proportionally increase with longer ttPORT with a greater fraction of LRR in the delayed group at 9 weeks ($P = .001$) and 10 weeks ($P < .001$), respectively.

We observed no statistically significant association between ttPORT and the rate of DM as the first event ($P = .87$). For cumulative incidence of MCCSD, where non-MCC death was treated as a competing risk, we observed a significant overall trend with increasing ttPORT (HR = 1.14 per 1-week increase, $P = .016$). However, when evaluating all causes of death, the overall trend in relation to 1-week increases in ttPORT, was not significant ($P = .13$).

Predictors of recurrence rates and time to PORT

To further explore potential contributing factors for the variation in ttPORT, we evaluated the incidence of complications of definitive surgery, skin grafts with and without complications, as well as staged surgical procedures (ie wide local excision and sentinel LN biopsy were not performed on the same date) (Table III). Skin grafts that had complications (sloughing, necrosis, etc.) were significantly associated with a longer ttPORT ($P < .001$).

Univariate analysis demonstrated no statistically significant associations between LRR and baseline clinical patient, tumor, and treatment characteristics, except for skin grafts with complications (HR: 6.3, $P < .001$) (Table IV). Given the limited LRR event

Table IV. Univariate associations with baseline characteristics and Loco-regional recurrences (N = 124)

Variable	HR	(95% CI)	P
ttPORT, per 1-wk increase (reference)	1.20	(1.07-1.36)	.002
Skin graft performed, with complication	6.32	(2.36-16.91)	<.001
Stage p/c-II (vs p/c-I)	2.15	(0.86-5.42)	.10
Clinical stage (vs path stage)	2.25	(0.80-6.34)	.13
Outside Institutions for RT (vs Institution 1)	2.75	(0.61-12.33)	.19
Size of primary, per 50% increase in size	1.20	(0.90-1.59)	.21
Female sex	0.55	(0.19-1.57)	.26
Head/neck primary	0.56	(0.18-1.71)	.31
T-stage T3/4 (vs T1/T2)	2.24	(0.45-11.05)	.32
LVSI	1.50	(0.54-4.16)	.44
Positive path margins	1.66	(0.45-6.11)	.44
No. of RT fractions, per 1-fraction increase	1.06	(0.89-1.25)	.52
Age, per 10-y increase	1.09	(0.72-1.67)	.67
Immunosuppressed	2.73	(0.90-8.29)	.077
Total RT dose, per 1-Gy increase	1.02	(0.91-1.13)	.77
PORT given to the elective draining LN basin	0.88	(0.30-2.60)	.82
Total RT dose ≥ 60 Gy (vs <60 Gy)	1.13	(0.32-3.96)	.85

This table includes all co-variables that were analyzed, regardless of whether or not there was statistical significance. Bold values signify statistical significance.

LN, Lymph node; LVSI, lymphovascular Invasion; PORT, post-operative radiation therapy; ttPORT, Time-to-PORT.

rate, a complete multivariable analysis to adjust the association between ttPORT and LRR for all potential confounders at once was not possible. Therefore, we adjusted the association between ttPORT and LRR by each variable listed as a sensitivity analysis to identify any potential cofounders. The association between ttPORT and LRR remained statistically significant after each adjustment except for skin grafts with complications.

DISCUSSION

To our knowledge, this study is the first to evaluate the relationship between the length of time interval from definitive surgery to the initiation of PORT with LRR rates for patients with localized MCC. We observed a 20% increase in the rate of LRR as the length of ttPORT increased by each additional week (HR: 1.20, $P = .002$).

Tsang et al¹⁸ reported a median delay of 11 weeks was associated with a 45% ($n = 5/11$) risk of LRR among 27 stage I-V MCC patients. Our study was designed to evaluate the impact of delay in initiating PORT to prevent LRR (rather than as a salvage or palliative treatment). Our results are consistent with other malignancies with similar repopulation and growth kinetics as MCC, such as head and neck squamous cell carcinoma, where worse locoregional control has been observed with prolonged ttPORT.¹²⁻¹⁶

We observed a significant worsening overall trend for increased MCCSD with increasing ttPORT ($P = .016$), suggestive that timely initiation of PORT is associated with a lower likelihood of dying with active disease and from an MCC-related event. However, there was no significant difference for death from all causes ($P = .13$). These results are consistent with a National Cancer Database study that found no association between ttPORT and OS in MCC patients.¹⁹

Additionally, ttPORT was a predictor of LRR despite the cohort of > 8 weeks receiving a slightly greater total dose (54 Gy vs 50 Gy, $P = .004$) and a greater proportion of patients that received PORT to the elective draining LN basin (45.2% vs 19.4%, $P = .008$), likely due to a greater proportion of patients who were clinically staged. This suggests that a higher total dose or a more generous field of radiotherapy may not be sufficient to reduce the increased risk of LRR.

The majority (82%, $n = 14/17$) of LRR occurred outside the radiation field (Table II). The low incidence of in-field treatment failures is in keeping with the documented efficacy of PORT and the intrinsic radiosensitivity of MCC.⁶⁻¹¹ Further, the higher rate of in-transit and regional recurrences with longer ttPORT suggests that delay in PORT may allow MCC cells within the surgical bed to spread, specifically along the lymphatic pathways. In other cancer types, nodal and DM can arise from distinct subclones, a possible basis for differential responses to adjuvant RT.²⁰

The role of elective RT to the pathologically uninvolved regional LN basins and in-transit

lymphatics is controversial for patients with localized MCC. For MCC arising in the limbs and trunk sites, comprehensive inclusion of these nodal basins is often prohibitive due to morbidity concerns or due to the treatment volume. For patients with H&N primary sites, or patients unable to successfully undergo pathological staging, we have an informed discussion with the patient to decide on inclusion of elective nodal coverage. There are also emerging reports regarding the abscopal effect of RT on MCC disease beyond the treated area, a process hypothesized to be related to increased antigen presentation and CD8+ T-cell recruitment.²¹⁻²⁴

While the findings of this analysis support the shortest possible duration of ttPORT for optimal locoregional control, it is generally advisable to allow a minimum of 2-3 weeks prior to initiation of PORT for sufficient wound healing, which can be location specific. This study uncovered that skin grafts with complications were associated with a prolonged ttPORT, and consequently increased LRR. Tarabdar et al reported on 188 stage I-IIIa MCC patients, where among the 140 patients who received PORT, the local control was excellent regardless of the size of the surgical margin (≤ 1 cm vs >1 cm; $P = .540$).²⁵ There was only 1 local failure in each group of 70 patients (wide vs narrow margins), concluding that if PORT is planned, narrower surgical margins are preferred as they reduce morbidity and in conjunction with our results might allow for timely initiation of PORT for optimal locoregional control.

Limitations of this study include the retrospective design and a relatively small number of patients. The latter must be regarded in the context of a rare cancer. While inclusion of patients treated at tertiary academic hospitals and community practices could add heterogeneity to treatment practice, it could also make these data more generalizable.

CONCLUSION

Prolonged time to initiation of PORT after definitive surgery is associated with poorer locoregional control and MCC-specific survival in patients with localized MCC. Each additional week delay was associated with a 20% increase in rate of LRR, and delay >8 weeks significantly increased the risk of LRR. These data suggest PORT should be initiated as early as feasible, ideally within 8 weeks of definitive surgery.

Conflict of interest

None disclosed.

REFERENCES

- Paulson KG, Park SY, Vandevean NA, et al. Merkel cell carcinoma: current US incidence and projected increases based on changing demographics. *J Am Acad Dermatol*. 2018;78(3):457-463.e2.
- Stang A, Becker JC, Nghiem P, Ferlay J. The association between geographic location and incidence of Merkel cell carcinoma in comparison to melanoma: an international assessment. *Eur J Cancer*. 2018;94:47-60.
- Becker JC, Stang A, DeCaprio JA, et al. Merkel cell carcinoma. *Nat Rev Dis Primers*. 2017;3:17077.
- Harms KL, Healy MA, Nghiem P, et al. Analysis of prognostic factors from 9387 Merkel cell carcinoma cases forms the basis for the new 8th edition AJCC staging system. *Ann Surg Oncol*. 2016;23(11):3564-3571.
- Lemos BD, Storer BE, Iyer JG, et al. Pathologic nodal evaluation improves prognostic accuracy in Merkel cell carcinoma: analysis of 5823 cases as the basis of the first consensus staging system. *J Am Acad Dermatol*. 2010;63(5):751-761.
- Petrelli F, Ghidini A, Torchio M, et al. Adjuvant radiotherapy for Merkel cell carcinoma: a systematic review and meta-analysis. *Radiother Oncol*. 2019;134:211-219.
- Strom T, Carr M, Zager JS, et al. Radiation therapy is associated with improved outcomes in Merkel cell carcinoma. *Ann Surg Oncol*. 2016;23(11):3572-3578.
- Lewis KG, Weinstock MA, Weaver AL, Otley CC. Adjuvant local irradiation for Merkel cell carcinoma. *Arch Dermatol*. 2006;142(6):693-700.
- Takagishi SR, Marx TE, Lewis C, et al. Postoperative radiation therapy is associated with a reduced risk of local recurrence among low risk Merkel cell carcinomas of the head and neck. *Adv Radiat Oncol*. 2016;1(4):244-251.
- Tseng YD, Nguyen MH, Baker K, et al. Effect of patient Immune status on the efficacy of radiation therapy and recurrence-free survival among 805 patients with Merkel cell carcinoma. *Int J Radiat Oncol Biol Phys*. 2018;102(2):330-339.
- Strom T, Naghavi AO, Messina JL, et al. Improved local and regional control with radiotherapy for Merkel cell carcinoma of the head and neck. *Head Neck*. 2017;39(1):48-55.
- Ang KK, Trotti A, Brown BW, et al. Randomized trial addressing risk features and time factors of surgery plus radiotherapy in advanced head-and-neck cancer. *Int J Radiat Oncol Biol Phys*. 2001;51(3):571-578.
- Mantravadi RV, Haas RE, Lieber EJ, Skolnik EM, Applebaum EL. Postoperative radiotherapy for persistent tumor at the surgical margin in head and neck cancers. *Laryngoscope*. 1983;93(10):1337-1340.
- Parsons JT, Mendenhall WM, Stringer SP, Cassisi NJ, Million RR. An analysis of factors influencing the outcome of postoperative irradiation for squamous cell carcinoma of the oral cavity. *Int J Radiat Oncol Biol Phys*. 1997;39(1):137-148.
- Peters LJ, Goepfert H, Ang KK, et al. Evaluation of the dose for postoperative radiation therapy of head and neck cancer: first report of a prospective randomized trial. *Int J Radiat Oncol Biol Phys*. 1993;26(1):3-11.
- Rosenthal DI, Liu L, Lee JH, et al. Importance of the treatment package time in surgery and postoperative radiation therapy for squamous carcinoma of the head and neck. *Head Neck*. 2002;24(2):115-126.
- National Comprehensive Cancer Network. Merkel cell carcinoma (version 1.2020), 2019. Accessed May 23, 2020. https://www.nccn.org/professionals/physician_gls/pdf/mcc.pdf
- Tsang G, O'Brien P, Robertson R, Hamilton C, Wratten C, Denham J. All delays before radiotherapy risk progression of Merkel cell carcinoma. *Australas Radiol*. 2004;48(3):371-375.
- Shinde A, Verma V, Jones BL, et al. The effect of time to postoperative radiation therapy on survival in resected Merkel cell carcinoma. *Am J Clin Oncol*. 2019;42(8):636-642.
- Naxerova K, Reiter JG, Brachtel E, et al. Origins of lymphatic and distant metastases in human colorectal cancer. *Science*. 2017;357(6346):55-60.
- Cotter SE, Dunn GP, Collins KM, et al. Abscopal effect in a patient with metastatic Merkel cell carcinoma following radiation therapy: potential role of induced antitumor immunity. *Arch Dermatol*. 2011;147(7):870-872.
- Schaub SK, Stewart RD, Sandison GA, et al. Does neutron radiation therapy potentiate an Immune response to Merkel cell carcinoma? *Int J Part Ther*. 2018;5(1):183-195.
- Xu MJ, Wu S, Daud AI, Yu SS, Yom SS. In-field and abscopal response after short-course radiation therapy in patients with metastatic Merkel cell carcinoma progressing on PD-1 checkpoint blockade: a case series. *J Immunother Cancer*. 2018;6(1):43.
- Bloom BC, Augustyn A, Pezzi TA, et al. Rescue of immunotherapy-refractory metastatic Merkel cell carcinoma with conventionally fractionated radiotherapy and concurrent pembrolizumab. *Front Oncol*. 2019;9:223.
- Tarabadkar ES, Fu T, Lachance K, et al. Narrow excision margins are appropriate for Merkel cell carcinoma when combined with adjuvant radiation: analysis of 188 cases of localized disease and proposed management algorithm. *J Am Acad Dermatol*. 2021;84(2):340-347.